VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
B.TECH. IV YEAR
CIVIL ENGINEERING

VII SEMESTER **R22**

Course Code	Title of the Course	L	T	P/D	CH	C
22PC1CE401	Foundation Engineering	3	0	0	3	3
22PC1CE402	Innovation and Entrepreneurship	3	0	0	3	3
	PROFESSIONAL ELECTIVE – III					
22PE1CE401	Earthquake Resistant Design of Buildings	3	0	0	3	3
22PE1CE402	Prestressed Concrete					
22PE1CE403	Advanced Structural Design					
22PE1CE404	Pre-Engineered Buildings					
22PE1CE405	Construction Technology and Project Management					
	PROFESSIONAL ELECTIVE – IV THROUGH SWAYAM-NPTEL					
	Geotechnical Engineering Stream	3	0	0	3	3
	Transportation Engineering Stream					
	Environmental Engineering Stream					
	Water Resources Engineering Stream					
	Geospatial Technologies Stream					
	OPEN ELECTIVE – III	3	0	0	3	3
22PC2CE411	Computer Aided Design Studio	0	0	2	2	1
22PC2CE306	Environmental Engineering Laboratory	0	0	2	2	1
22PW4CE401	Major Project Phase – I	0	0	6	6	3
Total		**15**	**0**	**10**	**25**	**20**

L – Lecture T – Tutorial P – Practical D – Drawing CH – Contact Hours/Week
C – Credits SE – Sessional Examination CA – Class Assessment ELA – Experiential Learning Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation LR – Lab Record
CP – Course Project PE – Practical Examination

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
B.TECH. IV YEAR
CIVIL ENGINEERING

VIII SEMESTER **R22**

Course Code	Title of the Course	L	T	P/D	CH	C
	PROFESSIONAL ELECTIVE – V					
22PE1CE406	Air Pollution and Control	3	0	0	3	3
22PE1CE407	Solid Waste Management					
22PE1CE408	Environmental Impact Assessment					
22PE1CE409	Industrial Hazardous Waste Management					
22PE1CE410	Waste Resource Recovery and Energy Systems					
	PROFESSIONAL ELECTIVE – VI					
22PE1CE411	Rock Mechanics	3	0	0	3	3
22PE1CE412	Ground Improvement Techniques					
22PE1CE413	Applications of Geo Synthetics					
22PE1CE414	Geo-Environmental Engineering					
22PE1CE415	Advanced Sub Surface Investigation and Instrumentation					
	OPEN ELECTIVE – IV	3	0	0	3	3
22PW4CE402	Major Project Phase – II	0	0	22	22	11
Total		**9**	**0**	**22**	**31**	**20**

L – Lecture T – Tutorial P – Practical D – Drawing CH – Contact Hours/Week
C – Credits SE – Sessional Examination CA – Class Assessment ELA – Experiential Learning Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation LR – Lab Record
CP – Course Project PE – Practical Examination

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC1CE401) FOUNDATION ENGINEERING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:
- To Plan Soil exploration programme for civil Engineering Projects
- To check the stability of slopes under various conditions
- To determine the lateral earth pressures to design retaining walls
- To predict the bearing capacity of shallow and deep foundation

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand the principles and methods of Geotechnical Exploration
CO-2: Evaluate the stability of slopes for different types of soil
CO-3: Calculate lateral earth pressures and check the stability of retaining walls
CO-4: Decide the suitability of shallow and deep foundations

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	3	3	2	2	3	2	2	2	1	3	3	3	3
CO-2	3	3	3	2	3	3	1	1	2	-	-	1	-	3	3
CO-3	3	3	3	2	3	-	-	-	2	-	1	-	-	3	3
CO-4	3	3	3	2	3	-	-	-	1	1	-	3	1	3	3

UNIT-I:
Subsoil Exploration: Methods of subsoil exploration - Direct, semi-direct, and indirect methods, Soundings by Standard, Dynamic, and Static Cone Penetration tests. Types of Boring, Types of soil sampler and sampling, Criteria for undisturbed sampling, planning of exploration programmes, Bore logs, Transport and preservation of samples, report writing.

UNIT-II:
Earth Slope Stability: Infinite and finite earth slopes – types of failures – factor of safety of infinite slopes – stability analysis by standard method of slices, Bishop's Simplified method– Taylor's Stability Number- Stability of slopes of earth dams under steady seepage, sudden drawdown, during and at the end of construction.

UNIT-III:
Earth Pressure: Types of Earth Pressure. Rankine's active and passive earth pressure, Smooth vertical wall with horizontal backfill. Coloumb's wedge theory.
Retaining Walls: Types of retaining walls – Principles and proportioning of retaining walls - Stability of retaining walls - Shallow and deep shear failure - Drainage of the backfill.

UNIT-IV:

Bearing Capacity: Safe and allowable bearing capacity, Terzaghi's, IS bearing capacity equations and its modifications for square, rectangular, and circular footings. General and local shear failure conditions. Factors affecting bearing capacity of soil. Allowable bearing pressure based on N-values and Plate load tests.

Shallow Foundations: Factors effecting locations of foundation and design considerations of shallow foundations, choice of type of foundations. Settlement analysis: Causes of settlement, Computation of settlement, allowable settlement. Measures to reduce settlement.

UNIT-V:

Pile Foundations: Types, Construction, load carrying capacity of single pile – Static and Dynamic formulae, Pile load tests, Load carrying capacity of pile groups, settlement of pile groups, Negative skin friction. Problems with foundations on expansive soils.

TEXT BOOKS:

1. Principles of Foundation Engineering, Braja M. Das, 8th Edition, Cengage, 2017
2. Soil Mechanics, Foundation Engineering, V. N. S. Murthy, CBS Publishers, 2018
3. Soil Mechanics and Foundation Engineering, K. R. Arora, 7th Edition, Standard Publishers Distributors, 2017

REFERENCES:

1. Soil Mechanics and Foundations, Muni Budhu, 3rd Edition, John Wiley & Sons, 2011
2. Foundation Analysis and Design, J. E. Bowles, 5th Edition, Tata McGraw-Hill Publishers, 2001
3. Geotechnical Engineering, S. K. Gulhati & Manoj Datta, Tata McGraw-Hill, 2017

ONLINE RESOURCES:

1. https://archive.nptel.ac.in/courses/105/105/105105176
2. https://archive.nptel.ac.in/courses/105/105/105105185

B.Tech. VII Semester

(22PC1CE402) INNOVATION AND ENTERPRENEURSHIP

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:
- To provide foundational knowledge about innovation, identifying innovators, and the process of Innovation Management
- To introduce innovation incubation, prototyping, and IPR Management emphasizing real-world examples and procedures
- To enable students to ideate and solve real-world problems using design thinking and engineering approaches
- To equip students with tools to analyse markets, understand customer needs, and validate innovative ideas
- To facilitate understanding of business model frameworks and strategies to convert technological innovations into viable business solutions

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Know about Innovation, types, challenges, and co-creation for Innovation
CO-2: Develop incubation process, prototyping, and generate IPR
CO-3: Improve the entrepreneurial mindset and skills for entrepreneurship
CO-4: Convert an innovation into a business solution by understanding markets and customers
CO-5: Design a business model and understand the fundamentals of business strategy

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	1	1	1	-	-	2	1	1	3	2	1	3	1	-	-
CO-2	2	1	1	-	1	-	-	2	3		2	2	1	-	-
CO-3	-	-	2	-	-	3		3	3	2	2	3	1	-	-
CO-4	2	-	2	1	1	2	3	3	3	2	3	2	1	-	-
CO-5	-	-	1	-	-	2	3	2	3	1	3	3	1	-	-

UNIT-I:
Introduction to Innovation & Innovation Management
Innovation and Creativity: Definitions and Key Differences, Role of Motivation and Current Business Trends in Innovation, Types of Innovation, Challenges of Innovation, Sources of Innovation, Idea Management System. Innovation Management: Innovation Life Cycle, Roles in Innovation Process, Experimentation, Participation & Co-creation for Innovation.

UNIT-II:

Innovation Incubation and Intellectual Property

Stages of Innovation Incubation: Idea validation, prototyping, market testing, scaling; Use Cases: Startups incubated in India and globally; Creation of Intellectual Property Rights, Types of IPR, Procedures for Patenting and Copyright Filing in India.

UNIT-III:

Introduction to Entrepreneurship: Entrepreneurship vs Innovation, Types of Entrepreneurship, Human side of entrepreneurship, Myths & Realities about entrepreneurship, Entrepreneurial qualities and how to develop them, Government Initiatives Promoting Entrepreneurship in India, Introduction to the Lean Startup methodology.

UNIT-IV:

Understanding Needs, Customers, and Markets: Divergent vs. Convergent Thinking, Identifying a problem, understanding needs Ideation: why and how, Developing Value Propositions and Validation Frameworks, Value Chain Analysis, How-Might-We Statements, Market Research: Segmentation, Impact, and Failures of Startups.

UNIT-V:

Business Model and Beyond: Business model, Business Model Canvas (BMC), Lean Canvas (LC), Product-Market Fit (PMF), Kano Model, Action Priority Matrix, Value-Ease Matrix, Go-to-Market (GTM) Strategies, Blue Ocean Strategy PESTLE Analysis, Porter's 5 forces model, Competitor landscape, SWOT analysis

TEXT BOOKS:

1. The Lean Startup: How Today's Entrepreneurs Use Continuous Innovation to Create Radically Successful Businesses, Eric Ries, Penguin Portfolio, 2011 (ISBN: 978-0670921607)
2. Entrepreneurship Simplified: From Idea to IPO, Ashok Soota, S. R. Gopalan, India Portfolio, 2021 (ISBN: 978-0143454915)
3. Entrepreneurship: New Venture Creation, David H. Holt, Pearson Education India, 2016 (ISBN: 978-9332568730)

REFERENCES:

1. Entrepreneurship, Robert D. Hisrich et al., McGraw-Hill, 11th ed., 2020 (ISBN: 978-9390113309)
2. The Manual for Indian Start-Ups, Vijay Kumar Ivaturi et al., Penguin Enterprise, 2020 (ISBN: 978-0143450856)
3. Engineered in India: From Dreams to Billion-Dollar Cyient, B. V. R. Mohan Reddy, Penguin Business, 2022 (ISBN: 978-0670097340)
4. Before You Start-Up: How to Prepare to Make Your Startup Dream a Reality, Pankaj Goyal, Fingerprint! Publishing, 2017 (ISBN: 978-8175994409)
5. How to Start Business in India: Idea to Implementation, Dibyendu Choudhury, Notion Press, 2023 (ISBN: 979-8890670267)

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1CE401) EARTHQUAKE RESISTANT DESIGN OF BUILDINGS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand seismology
- To explain and discuss single degree of freedom systems subjected to free and forced vibrations
- To acquire the knowledge of the conceptual design and principles of earthquake resistant designs as per IS codes
- To understand the importance of ductile detailing of RC structures

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Derive fundamental equations for SDOF systems in structural dynamics
CO-2: Discuss and explain causes and theories on earthquake, seismic waves, measurement of earthquakes
CO-3: Explain the importance of conceptual design and determine the base shear using IS methods
CO-4: Design and detail the beams and columns for ductile requirements as per IS provisions

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	1	1	-	-	-	-	-	1	-	-	1	1	1
CO-2	2	2	1	1	1	-	-	-	-	1	-	-	1	1	1
CO-3	1	1	3	2	3	2	2	1	1	1	-	1	1	2	3
CO-4	1	2	3	3	3	2	2	1	1	1	-	2	1	2	3

UNIT-I:
Introduction to Structural Dynamics: Elements of vibrating system – Degrees of freedom – Continuous system – Lumped mass idealization – Oscillatory motion – Free vibrations of single degree of freedom system – undamped, damped – Logarithmic decrement – Forced vibrations of single degree of freedom system – Harmonic loading

UNIT-II:
Earthquakes and Ground Motion: The interior of the earth – Causes of earthquakes – Nature and occurrence of earthquakes – Seismic waves – Effects of earthquake – Consequences of earthquake damage – Measurement of earthquakes - intensity and magnitude – Seismograph – Classification of Earthquakes – Seismic zonation

UNIT-III:

Conceptual Design: Introduction to functional planning – Continuous load path – Overall form – Simplicity, Uniformity and Symmetry – Elongated shapes – Stiffness and strength – Horizontal and vertical members – Twisting of buildings – Ductility - Flexible buildings

UNIT-IV:
Seismic Methods of Analysis: Basic assumptions – Principles in earthquake resistant designs – Seismic design requirements – Design earthquake loads – Permissible stresses – Seismic methods of analysis – Equivalent lateral force method – Dynamic analysis – Response spectrum method – Time history method -Numerical examples using static equivalent method

UNIT-V:
Ductile Design of Reinforced Concrete Buildings: Damage to RC buildings — Ductility considerations in earthquake resistant design of RC buildings – Impact of ductility – Requirements for ductility –Assessment of ductility – Factors affecting ductility – Ductile detailing considerations as per IS:13920 – Behaviour of beams, columns RC buildings during earthquake

TEXT BOOKS:
1. Earthquake Resistant Design of Structures, S. K. Duggal, Oxford University Press, 2013
2. Earthquake Resistant Design of Structures, Pankaj Agarwal and Manish Shrikhande, Prentice Hall of India, 2011
3. Dynamics of Structures, AK Chopra, 5th Edition, Pearson,2019

REFERENCES:
1. Structural Dynamics - Theory and Computation, Mario Paz, CBS Publisher, 6th Edition 2019
2. Earthquake Tips - Learning Earthquake Design and Construction, C. V. R. Murthy, IIT Kanpur and BMTPC, New Delhi, 2005

CODE BOOKS:
1. IS 1893 (Part 1): 2016 - Criteria for Earthquake Resistant Design of Structures – Part 1 : General Provisions and Buildings, Bureau of Indian Standards
2. IS 13920: 2016 - Ductile Design and Detailing of Reinforced Concrete Structures subjected to Seismic Forces – Code of Practice (First Revision), Bureau of Indian Standards

ONLINE RESOURCES:
1. https://archive.nptel.ac.in/courses/105/101/105101004/
2. https://archive.nptel.ac.in/courses/105/102/105102016/

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1CE402) PRESTRESSED CONCRETE

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:
- To describe the necessity of pre-stressed concrete structures and various techniques of pre-stressing
- To develop an understanding of various losses & deflections in prestressed members
- To understand the analysis of prestressed concrete members
- To develop an understanding of the analysis of composite members

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Acquire the knowledge of evolution of process of prestressing
CO-2: Explain various prestressing techniques
CO-3: Estimate the losses of prestress
CO-4: Analyse the prestressed concrete beams and check for deflection

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	-	-	-	-	-	-	1	-	-	-	-	-	1	1	1
CO-2	1	-	-	-	-	-	1	-	-	1	-	-	1	1	1
CO-3	1	2	2	2	2	1	2	1	1	1	-	1	1	2	2
CO-4	1	3	2	2	2	1	2	1	1	1	-	1	1	3	3

UNIT-I:
Introduction: Historic development – General principles of prestressing pre-tensioning and post- tensioning, Advantages and limitations of prestressed concrete, Materials – High strength concrete and high tensile steel, their characteristics, I.S. code provisions.

UNIT-II:
Prestressing Techniques: Methods and Systems of Pre-stressing, Pre-tensioning and post- tensioning methods, Analysis of post tensioning, Different systems of prestressing like Hoyer's system, Magnel system, Freyssinet system and Gifford-Udall system.

UNIT-III:
Losses of Prestress: Loss of prestress in pre-tensioned and post-tensioned members due to causes like Elastic shortening of concrete, Shrinkage of concrete, Creep of concrete, Relaxation of steel, Slip in anchorage, Frictional losses.

UNIT-IV:
Analysis of Sections for Flexure: Elastic analysis of concrete beams prestressed with straight, concentric, eccentric, bent and parabolic tendons.

UNIT-V:
Deflections of Prestressed Concrete Beams: Importance of control of deflections – factors influencing deflections – short term deflections of uncracked members, prediction of long-term deflections.
Composite Sections: Introduction – Analysis of stress – Differential shrinkage – General design considerations.

TEXT BOOKS:
1. Prestressed Concrete, Krishna Raju, 6th Edition, Tata McGraw-Hill 2018
2. Pre-stressed Concrete, N. Rajagopalan, 2nd Edition, Narosa Publications, 2005

REFERENCES:
1. Pre-stressed Concrete Structures, P. Dayaratnam, 6th Edition, Oxford & IBH Publishers 2018
2. Design of Pre-stressed Concrete Structures, T. Y. Lin & N. H. Burns, 3rd Edition, John Wiley & Sons, 1991
3. Pre-stressed Concrete Structures, M. K. Hurst, 2nd Edition, Tata McGraw-Hill, 2019
4. Pre-stressed Concrete, Ramamrutham, 5th Edition, Dhanpat Rai & Sons, 2013
5. Pre-stressed Concrete, K.U.Muthu, Agmil Ibrahim, Maganti Janardhana, M. Vijayanand, PHI, 2016

CODE BOOKS:
1. IS 1343:2012 (Reviewed in 2022), Prestressed Concrete – Code of Practice (Second Revision), Bureau of Indian Standards.

ONLINE RESOURCES:
1. https://nptel.ac.in/courses/105106117

B.Tech. VII Semester

(22PE1CE403) ADVANCED STRUCTURAL DESIGN

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:
- To understand the working stress / Limit state design principles and design the RC structures based on appropriate IS codes
- To identify the relevant IS codes and necessary codal provisions for the design of RC structural components / structures
- To determine the design shear forces and bending moments from the critical sections
- To sketch the reinforcement details in RC structures

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Apply the basic concepts of RCC for the design of complex RC structures
CO-2: Determine the various types of loads acting on RC structures
CO-3: Understand the various codal provisions for the design of structural components
CO-4: Design the RC structures for various loads and load combinations

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	1	1	2	1	1	1	1	-	1	1	1	1	2	1
CO-2	2	1	1	1	1	1	1	1	-	1	-	-	1	2	1
CO-3	1	2	1	1	2	1	1	1	-	1	-	-	1	2	1
CO-4	2	3	2	2	3	2	2	1	-	1	1	1	1	3	3

UNIT-I:
Combined Footings: Design of Rectangular Combined Footings, Design of Trapezoidal Combined Footings.

UNIT-II:
Staircases: Design of Dog legged Staircase - Stringer Beam, Waist Slab, Design of Open Newel Staircase

UNIT-III:
Concrete Bridges: IRC Class AA Tracked & Wheeled Vehicle loading, Design of deck slab bridge.

UNIT-IV:
Water Tanks: Design of RCC Water tanks resting on ground, Elevated Water tanks

UNIT-V:

Flat Slabs: Introduction, Proportioning of Flat slabs, Direct design method, Distribution of moments in column strips and middle strip, Moment and shear transfer from slabs to columns, Shear in Flat slabs, Check for One way and Two way shear.

TEXT BOOKS:

1. Advanced Reinforced Concrete Structures, Varghese, 2nd Edition, PHI, 2010
2. Reinforced Concrete Structures Vol – II, B. C. Punmia, Ashok Kumar Jain, Arun Kumar Jain, 5th Edition, Laxmi Publications, 2015
3. Advanced Reinforced Concrete Design, N. Krishna Raju, 3rd Edition, CBS Publishers, 2016

REFERENCES:

1. Essentials of Bridge Engineering, D. Johnson Victor, 6th Edition, Oxford and IBM Publication, 2019
2. Reinforced Concrete Design, S. U. Pillai, D. Menon, 4th Edition, Tata McGraw-Hill, 2021
3. Structural Design and Drawing: Reinforced Concrete and Steel, N. Krishna Raju, 4th Edition, Universities Press, 2022

CODE BOOKS:

1. IS 456: 2000 (Reviewed in 2021) - Plain and Reinforced Concrete - Code of Practice (Fourth Revision), Bureau of Indian Standards
2. IRC 112: 2020 - Code of Practice for Concrete Road Bridges
3. IS 3370: 2021 (Reviewed in 2022) – Concrete Structures for retaining Aqueous liquids - Code of Practice: Part 1 General Requirements (Second Revision)
4. IS 3370: 2021 – Concrete Structures for retaining Aqueous liquids - Code of Practice: Part 2 Plain and Reinforced Concrete (Second Revision)

ONLINE RESOURCES:

1. https://onlinecourses.nptel.ac.in/noc22_ce63/preview

Note: Question paper pattern for Final Examination

The end examination paper should consist of Part-A and Part-B. Part-A consists of two questions in Design and Drawing out of which one question is to be answered for 24 marks. Part-B consists of five questions on design out of which three are to be answered for 36 marks (12 x 3).

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1CE404) PRE-ENGINEERED BUILDINGS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:
- To distinguish between conventional steel buildings and PEB's
- To identify the Pre-Engineered Building components
- To estimate the loads on Pre-Engineered Buildings
- To design the various components of PEB frames

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand the functions of primary system, secondary system and bracing system of PEB components
CO-2: Calculate the dead, live, wind and seismic loads acting on PEB's
CO-3: Check the structural stability of PEB's
CO-4: Analyze and design the PEB system components

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	1	2	2	1	1	1	1	1	-	1	1	1	2	2	2
CO-2	1	2	2	1	2	1	1	1	-	1	1	1	1	2	2
CO-3	2	2	3	2	3	2	1	1	-	1	1	2	2	2	3
CO-4	2	3	3	2	3	1	1	1	-	1	1	2	2	2	3

UNIT-I:
Introduction to Pre-Engineered Buildings: Introduction – History - Advantages of PEB - Applications of PEB – Materials used for manufacturing of PEB - Differences between Conventional Steel Buildings and Pre-Engineered Buildings.

UNIT-II:
Pre-Engineered Building Components: Primary system - Main frames, Gable End frame, Secondary system - Purlins & Girts. Bracing System - Rod, Angle, Portal, Pipe bracing, Sheeting and Cladding - Roof Sheeting and Wall sheeting, Accessories - Turbo Ventilators, Ridge vents, Sky Lights, Louvers, Insulation, Stair cases.

UNIT-III:
Design Loads on Pre-Engineered Buildings: Design of PEB frame under the influence of Dead, Live, Collateral, Wind, Seismic and other applicable Loads - Serviceability limits as per code.
PEB Design Methodology: Design parameters of PEB frames - Depth of the section, Depth to Flange width ratios, Thickness of Flange to Thickness of Web ratios of sections as per IS code - Section sizes as per manufacturing limitations.

UNIT-IV:

Structural Stability System of PEB: Shear buckling effect (d/t ratio exceeding 67ε) - Effective cross-sectional area concept for compression members (d/t ratio exceeding 42ε) - Effect of d/t ratio for flexural members according to section classifications - Bracing system: Flange Bracing, Rod Bracing, Angle Bracing, Portal Bracing.

UNIT-V:

Analysis and Design of Pre-Engineered Buildings: Analysis and Design of Rigid Frames - Rigid Frame Moment Connection, Shear Connection - Anchor bolt and Base plate design (Pinned and Fixed).

TEXT BOOKS:

1. Pre-Engineered Steel Building, K. S. Vivek, P. Vyshnavi, Lambert Academic Publishing, 2017
2. Metal Building Systems: Design and Specifications, Alexander Newman, 3rd Edition, McGraw-Hill, 2014

REFERENCES:

1. Pre-Engineered Metal Building Iron Worker, Red-Hot Careers, Create Space Independent Publishing, 2018
2. Pre-Engineered Metal Building Systems, LAP Lambert Academic Publishing, 2023

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1CE405) CONSTRUCTION TECHNOLOGY AND PROJECT MANAGEMENT

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To learn the principles of construction methods
- To know the fundamentals of construction safety
- To understand various equipments used in construction
- To practice Project Planning

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Differentiate between different construction equipment
CO-2: Prepare, plan and control resources of projects
CO-3: Prepare construction schedule by using different methods
CO-4: Understand and apply ISO 9000 standards to the projects

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	1	2	-	1	1	-	-	-	1	1	1	1	1	1
CO-2	1	1	1	-	3	1	-	1	2	2	3	2	1	2	2
CO-3	2	1	1	-	3	1	-	1	2	2	3	2	1	2	2
CO-4	1	1	1	-	3	1	-	1	2	2	3	2	1	1	2

UNIT-I:
Fundamentals of Construction Technology: Introduction to construction Project, Construction activities, phases and processes, Construction schedule, Construction documents and records, Quality, Safety, Codes and regulations.

UNIT-II:
Construction Equipment: Earthwork, Piling, Concreting, Formwork, Fabrication and erection, Mechanized construction and productivity, Equipment economics, Excavators, Rollers, Dozers, Scrapers, Hoisting equipment - Cranes, draglines and clamshells, Concrete equipment.

UNIT-III:
Quality and Safety: Quality control, Quality Assurance, ISO 9000, Quality systems, Principles on safety - personnel, fire and electrical, Concept of green building and its practices.

UNIT-IV:

Construction Management: Tendering process, Construction Contracts, Construction planning techniques, Resource Planning.
Claim Management: Construction claims – causes and its process, Disputes - causes and their resolution methods, Project closure, Documentation

UNIT-V:
Networking Methods: Project scheduling, Fundamentals of network, Program Evaluation and Review Technique, Critical Path Method, Resource allocation methods. Introduction to Project Management Softwares

TEXT BOOKS:
1. Construction Technology, Subir K. Sarkar, Subhajit Saraswati, Oxford University Press, 2008
2. Construction Project Management Theory and Practice, Neeraj Jha, Pearson Education, 2015

REFERENCES:
1. Project Planning and Control with PERT and CPM, B. C. Punmia, K. K. Khandelwal, 4th Edition, Laxmi Publication, 2023
2. Construction Project Management, K. K. Chitkara, 4th Edition, McGraw-Hill, 2019
3. Construction Planning & Management, U. K. Srivastava, Galgotia Publications, 2000
4. Construction Planning Equipment and Methods, Peurifacy, Schexnayder, Sharpira, 10th Edition, Tata McGraw-Hill, 2024

ONLINE RESOURCES:
1. https://archive.nptel.ac.in/courses/105/104/105104161/
2. https://archive.nptel.ac.in/courses/105/103/105103093/

B.Tech. VII Semester

(22PC2CE411) COMPUTER AIDED DESIGN STUDIO

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To acquire knowledge on creating maps with GIS software, data entry and editing it
- To understand the application of remote sensing and GIS for real world applications
- To draw and modify the various building elements
- To learn the commands to visualize the models

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Create a new digitized map for a given topo sheet, entry of relevant data and editing it

CO-2: Analyse the geospatial data using attribute and spatial queries

CO-3: Design and manage comprehensive BIM models using Revit,

CO-4: Demonstrate proficiency in creating architecture, improve visualization and workflow

COURSE ARTICULATION MATRIX:

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	2	1	3	1	1	1	2	2	1	1	2	1	1
CO-2	2	3	2	3	3	1	1	1	2	2	1	1	1	3	2
CO-3	1	2	2	3	3	1	1	1	3	2	1	1	1	1	3
CO-4	1	2	2	3	3	1	1	1	3	2	1	1	1	1	3

LIST OF EXPERIMENTS / EXERCISES:

1. Introduction to GIS software.
2. Digital database creation – Point features, Line features, Polygon features.
3. Data Editing-Removal of errors – Overshoot and Undershoot, Snapping.
4. Data Collection and Integration, Non-spatial data attachment working with tables.
5. Dissolving and Merging, Clipping, Intersection and Union.
6. Buffering techniques, Spatial and Attribute query and Analysis.
7. DEM creation.

8. Introduction to Revit Interface, General drawing tools, templates, and AutoCAD file import.
9. Creating Levels, Grids, References of Plans, and Structural Components.
10. Drawing and Modifying Building Elements (Floor Plans, Walls, Doors, Windows, Curtain Walls) and Openings for the Structural Elements.
11. Creating Annotations and Documentation for Plans and Elevations.
12. Developing Advanced Modeling (Topography, Phasing).
13. Use Walkthroughs, Rendering and perspectives for Visualization.
14. Integration of BIM and GIS – A Case Study

Software:

Ex: 1-7 QGIS Software (Open Source)

Ex: 8-14 Building Information Modeling Software (Bentley pack)

TEXT BOOKS:

1. ArcGIS Pro 2.8., User guide.2021
2. QGIS User Manual
3. Learning ArcGIS Pro-2, Tripp Corbinn, 2nd edition, Packt publishing, 2020
4. AECOsim Building Designer
5. Bentley User Manual

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC2CE306) ENVIRONMENTAL ENGINEERING LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To develop proficiency in standard methods for water quality analysis and testing
- To demonstrate experimental techniques for wastewater characterization and treatment
- To master laboratory procedures for analyzing biological parameters in environmental samples
- To conduct air quality monitoring and analyze pollutant concentrations using modern equipment

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Execute standard procedures for physical, chemical, and biological analysis of water samples

CO-2: Analyze and interpret wastewater parameters including BOD, COD, solids content and other critical indicators

CO-3: Operate sophisticated laboratory equipment and follow quality control procedures for environmental testing

CO-4: Measure and assess air pollution parameters using standard sampling techniques and Equipment

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	3	3	2	3	2	2	2	1	2	-	3	3
CO-2	3	3	3	3	3	2	3	2	2	2	1	2	-	3	3
CO-3	3	2	2	3	3	2	2	3	2	2	2	2	-	3	3
CO-4	3	3	2	3	3	3	3	2	2	2	1	2	-	3	3

LIST OF EXPERIMENTS:

Experiments on Water Testing:

1. Examine the pH for the given water and soil samples.
2. Examine the Electrical Conductivity of water and soil samples.
3. Examine the Turbidity for the given water and soil samples.
4. Investigate the Hardness for the given water samples.

5. Investigate the Acidity for the given water samples.
6. Examine the Alkalinity for the given water samples.
7. Measure the Chloride content for the given water sample by Argentometric method.
8. Examination of Optimum Coagulant Dosage using Jar Test (Flocculation Test).

Experiments on Wastewater Testing:

9. Analyze Total Solids, Suspended Solids, and Dissolved Solids in wastewater samples using gravimetric methods
10. Measure Settleable Solids using Imhoff Cone method
11. Determine Dissolved Oxygen concentration using Winkler's Method
12. Analyze Biochemical Oxygen Demand (BOD) using 5-day incubation method
13. Measure Chemical Oxygen Demand (COD) using Open Reflux method

Experiments on Air Pollution Parameters:

14. Analyze Suspended Particulate Matter (SPM) and Respirable Suspended Particulate Matter (RSPM) using High Volume Air Sampler and PM10 Sampler

TEXT BOOKS:

1. Standard Methods for the Examination of Water and Wastewater, 24th Edition, APHA, 2023
2. Environmental Laboratory Analysis Methods Handbook, 5th Edition, 2022
3. Environmental Sampling and Analysis for Technicians, Maria Csuros, Taylor & Francis, 2018

REFERENCES:

1. Environmental Protection Agency (EPA) Digital Learning Resources, www.epa.gov/learning-resources
2. Modern Environmental Analysis Techniques, Maria Csuros, Taylor & Francis, 2016
3. Handbook of Environmental Laboratory Analysis, John Keith, 2022
4. Quality Assurance for Environmental Analysis, Philippe Quevauviller, 1995
5. Environmental Monitoring and Analysis, Randy D. Down, John Wiley & Sons, 2004

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1CE406) AIR POLLUTION AND CONTROL

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:
- To analyze sources, effects, and fundamentals of air pollution
- To evaluate air pollution monitoring, meteorology, and modeling techniques
- To examine control technologies for particulate & gaseous contaminants
- To assess control systems for gaseous pollutants and indoor air quality management

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Analyze air pollution sources and their effects on human health, environment and climate
CO-2: Design and implement air quality monitoring programs and apply appropriate modeling techniques
CO-3: Select and evaluate control technologies for particulate contaminants
CO-4: Design control strategies for gaseous pollutants and indoor air quality management

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	1	-		2	-	1	-	-	-	-	-	-	-	1	-
CO-2	-	-	2	-	-	2	2	-	-	-	-	-	-	1	-
CO-3	-	-	2	2	-	-	2	-	-	-	-	-	-	1	-
CO-4	-	-	2	2	-	-	1	-	-	-	-	-	-	1	-

UNIT-I:
Introduction: Air Pollution and its definition - Structure and composition of Atmosphere – Sources and classification of air pollutants - Effects of air pollutants on human health, vegetation & animals, Materials & Structures – Effects of air Pollutants on the atmosphere, Soil & Water bodies – Long- term effects on the planet – Global Climate Change, Ozone Holes – Ambient Air Quality and Emission Standards – Air Pollution Indices – Emission Inventories – Ambient and Stack Sampling and Analysis of Particulate and Gaseous Pollutants - Factors influencing air pollution.

UNIT-II:
Air Pollution Modelling & Monitoring: Meteorology – Wind roses – lapses rates – mixing depth atmospheric dispersion – plume behaviour accumulation, estimation of pollutants – Effective stack height- Transport & Dispersion of Air Pollutants – Modelling Techniques – Air Pollution Climatology. stack emission standards, stack emission monitoring, ambient air quality monitoring, ambient air quality survey.

UNIT-III:

Control of Particulate Contaminants: Factors affecting Selection of Control Equipment – Gas Particle Interaction, – Working principle of Gravity Separators (cyclone), Centrifugal separators Fabric filters, Particulate Scrubbers, Electrostatic Precipitators – Operational Considerations - Process Control and Monitoring – Costing of APC equipment – Case studies for stationary and mobile sources.

UNIT-IV:
Control of Gaseous Contaminants: Factors affecting Selection of Control Equipment– Working principle of absorption, Adsorption, condensation, Incineration, Bio scrubbers, Bio filters – Process control and Monitoring - Operational Considerations - Costing of APC Equipment – Case studies for stationary and mobile sources.

UNIT-V:
Indoor Air Quality Management: Source types and control of indoor air pollutants, sick building syndrome types – Radon Pollution and its control – Membrane process - UV photolysis – Internal Combustion Engines - Sources and Effects of Noise Pollution – Measurement – Standards –Control and Preventive measures.

TEXT BOOKS:

1. Air Pollution Science Engineering and Management Fundamentals, Khare M, Sharma P, Kota S.H, Sumanth C, ISBN 9780367750527, CRC Press, 2024
2. Air Pollution and Control, K.V.S.G. Murali Krishna, Environmental Protection Society, 2016

REFERENCES:

1. Air Pollution Control: Design Approach, Cooper & Alley, 4th Edition, 2010
2. Indoor Air Pollution Control, Thad Godish, 1st Edition, 2019
3. Air Pollution Control in Indian Context, TERI Publications
4. CPCB Guidelines and Standards for Air Quality Management
5. National Clean Air Programme: Implementation Guidelines

ONLINE RESOURCES:

1. https://archive.nptel.ac.in/courses/105/107/105107213/
2. https://www.edx.org/learn/pollution/imt-air-pollution-causes-and-impacts
3. Air Pollution – a Global Threat to our Health | Coursera

MODELING TOOLS:

1. AERMOD Cloud Platform (aermod.com)
2. CALPUFF Modeling system (calpuff.com)
3. EPA's Air Quality Models (epa.gov/scram)

DATA RESOURCES:

1. realtime.cpcb.gov.in (CPCB Air Quality Data)
2. aqicn.org (Global Air Quality Index)
3. openaq.org (Open Air Quality Data)

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1CE407) SOLID WASTE MANAGEMENT

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To analyze waste management systems and circular economy principles within regulatory frameworks
- To evaluate waste characterization techniques and develop resource recovery strategies
- To design efficient collection, transportation, and reverse logistics systems
- To assess sustainable waste processing technologies and disposal methods in circular economy context

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Design integrated solid waste management systems incorporating circular economy principles and regulatory requirements
CO-2: Implement waste characterization and develop resource recovery strategies using modern technologies
CO-3: Optimize waste collection, storage, and reverse logistics systems for maximum resource recovery
CO-4: Evaluate and select appropriate processing technologies and disposal methods aligned with circular economy goals

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	1	1	-	-	-	-	-	-	-	-	-	-	1	-
CO-2	-	2	1	-	-	2	2	-	-	-	-	-	-	1	-
CO-3	-	1	2	1	-	-	-	-	-	-	-	-	-	1	-
CO-4	-	1	2	-	-	1	1	-	-	-	-	-	-	1	-

UNIT-I:
Sources, Classification, Regulatory Framework and Circular Economy Principles: Types and Sources of solid and hazardous wastes-Need for solid and hazardous waste management- Circular Economy fundamentals and waste hierarchy-Salient features of Indian legislations on management and handling of various wastes-Elements of integrated waste management and circular economy approaches-Roles of stakeholders in circular waste management-Financing and Public Private Participation-Extended producer responsibility in circular economy

UNIT-II:
Waste Characterization and Resource Recovery: Waste generation rates and variation-Composition, physical, chemical and biological properties-Hazardous Characteristics and TCLP tests-Waste sampling and characterization plan-Circular

design principles and material selection-Source reduction and waste prevention strategies-Waste exchange platforms and industrial symbiosis-Extended producer responsibility implementation-Design for recycling and reuse

UNIT-III:
Storage, Collection, Transport and Reverse Logistics: Handling and segregation at source-Storage and collection systems for resource recovery- Analysis of Collection systems and optimization-Reverse logistics and material recovery-Transfer stations and waste allocation-Smart technologies in waste collection-Compatibility and handling of hazardous wastes-Digital tracking and monitoring systems

UNIT-IV:
Resource Recovery and Processing Technologies: Objectives of waste processing in circular economy-Material separation and recovery technologies- Biological and chemical conversion technologies-Composting and bio-conversion methods-Thermal conversion and energy recovery-Value addition to recovered materials-Smart processing technologies-Digital solutions in waste processing

UNIT-V:
Sustainable Disposal and Material Recovery: Waste disposal in circular economy context-Modern landfill design and operation-Resource recovery from landfills-Landfill mining technologies-Leachate and gas management for resource recovery-Environmental monitoring systems-Rehabilitation Strategies-Future trends in waste management

TEXT BOOKS:
1. Integrated Solid Waste Management, Engineering Principles and Management Issues, Tchobanoglous G, Theisen H, Vigil SA, McGraw-Hill Education, Indian Edition, 2014
2. Waste Management Practices: Municipal, Hazardous and Industrial, John Pichtel, 2nd Edition, CRC Press, 2014
3. Prospects and Perspectives of Solid Waste Management, B.B. Hosetti, 1st Edition, New Age International Publishers, 2006

REFERENCES:
1. Solid Waste Engineering, Vesilind PA, Worrell W, Reinhart D, 2nd Edition, Brooks/Cole Thomson Learning Inc., 2010
2. Environmental Engineering, Peavy H.S., Rowe D.R., Tchobanoglous G., 1st Indian Edition, McGraw-Hill Education, 2017

ONLINE RESOURCES:
1. https://archive.nptel.ac.in/courses/105/103/105103205/
2. WBGx: Solid Waste Management | edX
3. Municipal Solid Waste Management in Developing Countries | Coursera
4. http://cpheeo.gov.in/cms/manual-on-municipal-solid-waste-management-2016.php

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1CE408) ENVIRONMENTAL IMPACT ASSESSMENT

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To analyze the principles and methodology of Environmental Impact Assessment
- To evaluate various impact identification and prediction techniques
- To assess environmental impacts of developmental activities across different sectors
- To develop environmental management plans and audit protocols

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Prepare comprehensive Environmental Impact Assessment reports following regulatory guidelines
CO-2: Apply appropriate methodologies for impact identification and prediction
CO-3: Conduct environmental impact studies for various developmental projects
CO-4: Design environmental management plans and perform environmental audits

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	1	2	3	3	-	3	3	-	1	3	-	2	1	3	2
CO-2	1	2	3	3	-	3	3	-	1	-	3	2	1	3	3
CO-3	1	2	3	3	-	3	3	-	1	-	3	2	1	3	3
CO-4	1	2	3	3	3	3	3	-	2	3	3	3	2	3	3

UNIT-I:
Introduction: Introduction to Environmental Impact Assessment (EIA), Historical development of EIA, Definition of EIA and EIS, Types and limitations of EIA, Elements of EIA, Terms of reference in EIA, Classification of Environmental parameters, Initial Environmental Examination, Preparation of Environmental Base map, EIA procedure, Preparation of EIS, Public Participation in EIA, Environmental Clearence.

UNIT-II:
Impact Identification and Prediction Methodologies: Introduction, criteria for selection of EIA Methodology, EIA methods - Adhoc method, Check List Method, Matrix method, Network method, Environmental media quality Index method, Overlays method, Cost benefit analysis method, Expert systems in EIA. Prediction tools for EIA.

UNIT-III:
Assessment of Impact on Developmental Activities: Assessment of Impact of developmental Activities on soil, ground water, surface water, vegetation, air, wild life, Noise & Socio-Economic Environment, Environmental impacts of Deforestation-Causes and effects.

UNIT-IV:

Environmental Audit: Environmental Audit-Guidelines, Types of environmental Audit, Audit protocol, Stages of Environmental Audit onsite activities, evaluation of audit data and preparation of Audit Report, Post Audit Activities, Case studies On EIA, Environmental laws and regulations- Air act, Water act, environmental protection act.

UNIT-V:

Environmental Management Plan: Environmental Management Plan - preparation, implementation and review – Mitigation and Rehabilitation Plans – Policy and guidelines for planning and monitoring programmes – Post project audit – Ethical and Quality aspects of Environmental Impact Assessment- Case Studies.

TEXT BOOKS:

1. Environmental Impact Assessment: Theory and Practice, Dr. M. Anji Reddy, BS Publications, 2023
2. Methods of Environmental and Social Impact Assessment, Therivel & Wood, 4th Edition, 2017
3. Advanced Introduction to Environmental Impact Assessment, Morrison-Saunders, 2023

REFERENCES:

1. Environmental Impact Assessment Methodologies, Anjaneyulu Y, Manickam V, B.S. Publications, Hyderabad, 2023
2. EIA Guidelines and Notification Updates, MoEF & CC
3. Environmental Clearance Process in India: A Comprehensive Guide
4. Sector-specific EIA Guidelines

ONLINE RESOURCES:

1. https://archive.nptel.ac.in/courses/124/107/124107160/

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1CE409) INDUSTRIAL HAZARDOUS WASTE MANAGEMENT

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand the concepts of hazardous waste management
- To apply the principle of waste characterization, storage, transport and processing
- To understand the principles and concepts nuclear, e-waste waste management
- To analyze the process of secured landfills and Hazardous waste management

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand the fundamental concepts of hazardous waste management
CO-2: Use the knowledge to resolve the problems on storage, transport and processing
CO-3: Apply the knowledge to resolve the practical problems related to nuclear, e-waste waste Management
CO-4: Impart the gained knowledge and skills in Hazardous waste management

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	1	-	-	-	2	2	-	-	-	1	2	-	3	2
CO-2	1	3	2	2	1	2	2	-	-	-	1	1	-	3	3
CO-3	-	1	3	2	2	2	2	1	-	-	1	-	-	3	3
CO-4	-	-	-	1	-	-	-	2	2	2	3	3	-	2	3

UNIT-I:
Introduction to Hazardous waste management: Need for hazardous waste management – Sources of hazardous wastes – Effects on community – classification – Storage and collection of hazardous wastes – Problems in developing countries – Protection of public health and the environment.

UNIT-II:
Waste Characterization, Storage, Transport and Processing: Hazardous Waste Characterization– Hazardous waste inventory- Source reduction of hazardous wastes - Handling and storage of Hazardous wastes –Waste Compatibility Chart– Hazardous waste treatment technologies – Physical, chemical and thermal treatment of hazardous waste – Solidification – Chemical fixation – Encapsulation – Incineration.

UNIT-III:
Nuclear, E-waste Management: Nuclear Waste-Characteristics and types Uranium mining and processing wastes-Power reactor wastes-Spent fuel management-Health and environmental effects

E-Waste Management: Characterization of e-waste-Collection and segregation methods-Recycling and recovery technologies-Precious metal recovery Extended producer responsibility-E-waste regulations and guidelines

Battery Waste Management: Types of batteries and their composition-Collection systems-Storage and handling protocols-Recycling technologies-Metal recovery processes-Battery waste regulations-Safe disposal methods

UNIT-IV:

Management of Hazardous Wastes: Identifying a hazardous waste – methods – Quantities of hazardous waste generated – Components of a hazardous waste management plan – Hazardous waste minimization – Disposal practices in Indian Industries– Future challenges - Emergency Response - National Response Team and Regional Response Teams; National Contingency Plan and Regional Contingency Plans; National Response Center; State, Local and Industry Response Systems, Secured Landfill.

UNIT-V:

Secure Landfills: Hazardous waste landfills – Site selections – landfill design and operation – Regulatory aspects – Liner System- Liners: clay, geomembrane, HDPE, geonet, geotextile – Cover system- Leachate Collection and Management – Environmental Monitoring System- Landfill Closure and post closure care - Underground Injection Wells.

TEXT BOOKS:

1. Hazardous Waste Management: Contemporary Challenges and Solutions, Michael D. LaGrega, 2nd Edition, 2015
2. Handbook of Industrial and Hazardous Wastes Treatment, Lawrence K. Wang, Yung-Tse Hung, Howard Lo, Constantine Yapijakis, 2017
3. Electronic Waste Management and Treatment Technology, M. N. V. Prasad, Majeti, 2019

REFERENCES:

1. Basic Hazardous Waste Management, William C. Blackman Jr., 3rd Edition, 2016
2. Nuclear Waste Management: Science, Technology, and Policy, Man-Sung Yim, 2021

ONLINE RESOURCES:

1. https://archive.nptel.ac.in/courses/105/106/105106056/

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1CE410) WASTE RESOURCE RECOVERY AND ENERGY SYSTEMS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand the concepts of energy from waste
- To understand the principle and process of thermal conversion technology (TCT)
- To understand the principle and process of chemical and biological conversion technology (CCT & BCT)
- To understand the principles and processes of biomass energy technology (BET) and conversion process and devices (P&D) for solid wastes

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Apply the fundamental concepts of energy from waste
CO-2: Apply the acquired knowledge to resolve the practical problems on TCT and CCT
CO-3: Apply the knowledge to resolve the practical issues on BCT and BET
CO-4: Understand the principle of the LCA and GHG accounting to adopt mitigation strategies

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	2	2	3	3	2	1	2	2	2	-	3	3
CO-2	3	3	3	2	3	3	3	2	2	2	2	2	-	3	3
CO-3	3	3	3	2	3	3	3	2	2	2	2	2	-	3	3
CO-4	3	3	3	3	3	3	3	3	2	3	3	3	-	3	3

UNIT-I:

Introduction to Energy from Waste: Classification of waste as fuel—agro-based, forest residue, industrial waste, MSW—conversion devices—incinerators, gasifiers, digesters, Environmental monitoring system for landfill gases, Environmental impacts; measures to mitigate environmental effects due to incineration.

UNIT-II:

Thermal and Chemical Conversion Technologies: Fundamentals of thermal processing – combustion system – pyrolysis system – gasification system – environmental control system – energy recovery system – incineration. Acid & Alkaline hydrolysis – hydrogenation; solvent extraction of hydrocarbons; solvolysis of wood; bio-crude; biodiesel production via chemical process; catalytic distillation; transesterification methods; Fischer-Tropsch diesel.

UNIT-III:
Biological Conversion Technologies: Nutritional requirement for microbial growth – types of microbial metabolism – types of microorganisms – environmental requirements – aerobic biological transformation – anaerobic biological transformation – aerobic composting – low solid anaerobic digestion – high solid anaerobic digestion – development of anaerobic digestion processes and technologies for treatment of the organic fraction of MSW – Biodegradation and biodegradability of substrate; biochemistry and process parameters of biomethanation - other biological transformation processes.

UNIT-IV:
Biomass Energy Technologies: Biomass energy resources – types and potential; Energy crops - Biomass characterisation (proximate and ultimate analysis); Biomass pyrolysis and gasification; Biofuels – biodiesel, bioethanol, Biobutanol; Algae and biofuels; Pellets and bricks of biomass; Biomass as boiler fuel; Social, economic and ecological implications of biomass energy.

UNIT-V:
Life Cycle Assessment and Carbon-Footprint: Phases of Life Cycle Assessment: Goal and Scope Definition - Life Cycle Inventory - Life Cycle Impact Assessment – Interpretation - LCA Software - SimaPro Software- Sensitivity Analysis - Carbon-Footprint: Introduction to GHG emission and Inventorization – Scope 1, Scope 2 and Scope 3 Emission-Decarbonization Approach-Emissions Inventorization-Target Setting-Mitigation Strategies

TEXT BOOKS:
1. Waste-to-Energy: Technologies and Project Implementation, 4th Edition, Marc J. Rogoff, Francois Screve, 2011
2. Waste-To-Energy Technologies and Global Applications, Efstratios N. Kalogirou, CRC Press, 2024
3. Waste to Energy Efficient Municipal Solid Waste Management, Dr. Ranjita Roy Sarkar, 2023

REFERENCES:
1. Waste to Energy in the Age of the Circular Economy: Compendium of Case Studies and Emerging Technologies, Asian Development Bank, 2020
2. Waste-to-Energy Approaches Towards Zero Waste: Interdisciplinary Methods of Controlling Waste, Johannes Fink, 2017

ONLINE RESOURCES:
1. https://archive.nptel.ac.in/courses/103/107/103107125/

B.Tech. VIII Semester

(22PE1CE411) ROCK MECHANICS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand the concepts of Rock Mechanics and various terminology involved
- To conduct experiments as well as to analyze and interpret data related to rock mechanics
- To impart an understanding of the basic principles, and latest developments in real-world problems
- To connect theoretical knowledge to real-life problems

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand the classification of rocks and its properties
CO-2: Determine different engineering properties of rock
CO-3: Relate to the latest trends, modern standards, and state-of-the-art techniques for understanding rock mechanics and engineering
CO-4: Predict the mode of failure of rock structures and implement appropriate preventive measures

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	1	1	1	1	-	-	-	-	1	3	3	3	1	1
CO-2	3	3	2	3	2	2	-	-	-	1	3	2	3	2	3
CO-3	3	3	3	3	3	1	-	-	2	-	3	2	1	3	2
CO-4	3	3	3	3	3	2	1	-	3	1	2	3	-	2	1

UNIT-I:

Introduction to Rock Mechanics: Development of rock mechanics - problems of rock mechanics - applications and scope of rock mechanics - Classification by Rock Quality Designation - Rock structure Rating - Geomechanics and NGI classification systems.

UNIT-II:

Laboratory Testing: Rock sampling - Determination of density - Porosity and Water absorption - Uniaxial Compressive strength - Determination of elastic parameters - Tensile strength - Shear Strength - Flexural strength - Strength criterion in rocks - Swelling and slake durability - Permeability - Point load strength - Dynamic methods of testing - Factors affecting strength of rocks.

UNIT-III:
In-situ Testing: Necessity and Requirements of in - situ tests - Types of in - situ tests - Flat jack Technique - Hydraulic Fracturing Technique - Plate Load Test - Shear Strength Test - Radial Jack Test - Goodman Jack Test and Dilatometer Test.

UNIT-IV:
Methods of Improving Rock Mass Properties: Rock Reinforcement - Rock bolting - Mechanism of Rock bolting - Principles of design - Types of rock bolts. Pressure grouting - grout curtains and consolidation grouting.

UNIT-V:
Geostructures on Rock mass:
Slopes: Causes of landslides - Modes of failure - Methods of analysis - Prevention and control of rock slope failure - Instrumentation for Monitoring and Maintenance of Landslides.
Foundations: Shallow foundations - Pile and well foundations - Basement excavation - Foundation construction - Allowable bearing pressure.
Tunnels: Rock stresses and deformation around tunnels - Rock support interaction - Tunnel driving methods - Design of tunnel lining.

TEXT BOOKS:
1. Introduction to Rock Mechanics, Goodman, 2nd Edition, Wiley, 1989
2. Engineering in Rocks for Slopes, Foundations and Tunnels, Ramamurthy T., 3rd Edition, Prentice Hall of India, 2014
3. Fundamentals of Rock Mechanics, Jaeger J. C. and Cook N. G. W., 3rd Edition, Chapman and Hall, 1979

REFERENCES:
1. Underground Excavation in Rock, Hoek E. and Brown E. T., 1982
2. Rock Mechanics for Underground Mining, Brady B. H. G. and Brown E. T., Chapman & Hall, 3rd Edition, Springer Science & Business Media, 2007
3. Engineering Rock Mass Classification, Singh Bhawani, Goel R. K., Elsevier, 2011

ONLINE RESOURCES:
1. https://nptel.ac.in/courses/105107208
2. https://onlinecourses.nptel.ac.in/noc22_ce90

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1CE412) GROUND IMPROVEMENT TECHNIQUES

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:
- To learn various ground improvement techniques
- To understand methods of compaction for ground improvement
- To explain various physical and chemical modifications for ground improvement
- To choose the foundation and/or treatment method based on the site condition

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand different techniques of ground improvement techniques based on the various types of in-situ soils
CO-2: Select the suitable and economical ground improvement technique which is for soil strengthening
CO-3: Design and construction of reinforced earth structures
CO-4: Apply knowledge of ground improvement techniques for mechanical soil stabilization

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	3	1	-	-	3	1	-	-	1	3	-	2	1
CO-2	3	3	3	2	-	-	2	-	-	-	2	2	-	3	3
CO-3	3	3	3	3	1	-	2	-	-	-	2	3	-	3	2
CO-4	3	3	3	1	1	-	2	-	1	-	2	3	-	3	2

UNIT-I:
Introduction to Ground Modification: In-situ densification methods- Granular soils - Vibration and Impact at the Ground surface and depth. Cohesive soils-Preloading, Dewatering, Drain wells, Sand drains, Sandwich geo-drains, Prefabricated Vertical drains (PVD), Vacuum consolidation, Stone columns, Lime columns, Thermal methods.

UNIT-II:
Mechanical Stabilization: Soil aggregate mixtures, Properties and proportioning techniques, soft aggregate stabilization, compaction, Field compaction control.
Expansive Soils: Problems in Expansive soils, Mechanism of swelling, swell pressure, swell potential, Heave, Tests for identification, , Foundation techniques in Expansive soils.

UNIT-III:
Chemical Stabilization: Cement and lime - Mechanism, Factors affecting and properties, Construction techniques. Uses of modern additives.

UNIT-IV:

Reinforced Earth: Principles, Components of reinforced earth walls, Factors governing the design of reinforced earth walls, Design principles of reinforced earth walls. Introduction to geotextiles and geomembranes, Functions, and applications of geotextiles.

UNIT V:

Grouting and Nailing: Mechanism, Factors affecting, Construction methods. Grouting materials, Soil nailing.

TEXT BOOKS:

1. Ground Improvement Techniques, Dr. G. V. R. Purshotham Raj, Laxmi Publications, 4th Edition, 2006
2. Ground Improvement Techniques, Bikash Chandra Chattopadhyay, Joyanta Maity, Prentice Hall of India, 2018
3. Designing with Geosynthetics, Robert M. Koerner, Vol. 1, 6th Edition, 2012

REFERENCES:

1. Ground Improvement, Moseley M. P., Blackie Academic and Professional, Boca Taton, 2nd Edition, 2004
2. Ground Control and Improvement, Xanthakos P. P., Abramson L. W., Brucwe D. A., John Wiley and Sons, 1994
3. Principles and Practice of Ground Improvement, Jie Han, Wiley India, 2018

ONLINE RESOURCES:

1. https://archive.nptel.ac.in/courses/105/108/105108075
2. https://archive.nptel.ac.in/courses/105/105/105105210

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1CE413) APPLICATIONS OF GEO SYNTHETICS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To create awareness of the latest trends, modern standards, and state of the art techniques for solving geotechnical engineering problems
- To develop an ability to design a Geo-synthetic system to meet desired needs such as economic, environmental, and sustainability related
- To identify the latest trends in practice in numerous special aspects of civil engineering
- To apply the basic knowledge and to solve critical civil engineering problems in the field

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand different functions and applications of geo synthetics in Civil Engineering

CO-2: Identify the critical awareness of current issues in Geo environmental Engineering

CO-3: Interpret various techniques, and skills for application of geo synthetics in pavement construction

CO-4: Solve various Geotechnical Engineering problems using geo-synthetics

COURSE ARTICULATION MATRIX:

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	2	1	-	-	-	2	1	2	3	-	2	1
CO-2	3	3	3	1	1	1	2	-	1	1	3	2	-	3	1
CO-3	3	3	3	2	1	-	1	-	-	1	3	3	-	3	2
CO-4	3	3	3	3	1	-	3	1	2	2	3	3	-	3	3

UNIT-I:

An Overview of Geo-synthetics: Classification of Geo-synthetics, Functions and Applications, Raw Materials Used, Manufacturing, Sustainability.

Geo synthetic Properties and Testing: Physical properties - Specific Gravity, Mass per unit area, Thickness, Stiffness, POA, AOS. Mechanical properties – Tests on tensile strength, burst strength, tear, impact, Puncture, friction behavior, pull-out, seam strength, and compressibility. Hydraulic properties - In plane & Cross plane permeability, Long-term flow test. Endurance properties - Abrasion, UV-Degradation,

UNIT-II:

Soil Reinforcement: Mechanism, Reinforced slopes, Embankments on soft ground, Reinforced Embankments and Reinforced soil walls, Gabion walls, Internal and External Stability, Slope stabilization.

UNIT-III:
Geosynthetics for Highways: Roadway Reinforcement, Separation-Filtration-Drainage in highways: Mechanism, Applications, Functions, Reflection cracking, Pavement rehabilitation and repair, Widening of existing road embankments.

UNIT-IV:
Ground Improvement: Dewatering Systems: Prefabricated Vertical drains (PVD) designs, Sand Drains and French Drains, Bearing capacity: Geocell reinforced sand overlaying soft clay.

UNIT-V:
Geo-environmental Applications: Geomembranes for landfills and ponds, Geosynthetic clay liners (GCLs), Filtration, drainage and erosion control, Slope protection, and Encapsulation. Moisture barriers.

TEXT BOOKS:

1. Designing with Geo-synthetics, R. M. Koerner, 6th Edition, Xlibris Corporation, 2012
2. Engineering with Geo-synthetics, R. G. Venkatappa, R. G. V. S. Suryanarayana, Tata McGraw-Hill, 1990
3. An Introduction to Soil Reinforcement and Geosynthetics, G. L. Sivakumar Babu, University Press, 2005

REFERENCES:

1. Ground Improvement, Mosley, 2nd Edition, CRC Press, 2004
2. Earth Reinforcement and Soil Structures, C. J. F. P. Jones, Elsevier, 2013
3. Engineering Principles of Ground Modifications, M. R. Hausmann, McGraw-Hill, 1990
4. Ground Control and Improvement, Xanthakos, Abramson, Bruce, John Wiley & Sons, 1994
5. A Guide to Geotextiles Testing, J. N. Mandal, D. G. Divshikar, New Age International, 2002

ONLINE RESOURCES:
1. https://archive.nptel.ac.in/courses/105/106/105106052
2. https://nptel.ac.in/courses/105101143

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1CE414) GEO-ENVIRONMENTAL ENGINEERING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand various sources of contamination of ground and to characterize contaminated ground
- To apply the knowledge of Geo environmental engineering on problems associated with soil contamination
- To explain the extent of contamination and arrive at a safe disposal of waste
- To identify and remediate the contaminated soils by different techniques thereby protecting the environment

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand Soil-environment interaction and mechanisms of soil-water interaction
CO-2: Learn about groundwater flow and predict contaminant transport phenomenon
CO-3: Create an idea in the development of various landfill design techniques
CO-4: Apply remediation techniques for contaminated site

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	3	3	2	-	3	-	-	-	1	1	-	2	1
CO-2	3	3	3	3	3	1	3	-	-	-	-	2	-	1	1
CO-3	3	3	3	2	3	1	3	-	-	-	-	2	-	3	3
CO-4	3	3	3	2	3	1	3	-	-	-	-	3	-	2	3

UNIT-I
Generation of Wastes and Consequences of Soil Pollution: Introduction to Geo environmental engineering - Environmental cycle - Sources, production, and classification of waste - Causes of soil pollution - Effect of contaminant on subsoil - Factors governing soil pollution and interaction with clay minerals.

UNIT-II:
Site Selection and Safe Disposal of Waste: Safe disposal of waste - Site selection for landfills -Characterization of landfill sites and waste Risk assessment - Stability of landfills - Current practice of waste disposal - Monitoring facilities - Passive containment system - Application of geosynthetics in solid waste management in landfills - Rigid or flexible liners.

UNIT-III:
Contaminant Transport: Contaminant transport in Subsurface: Advection, Diffusion, Dispersion - governing equations - Contaminant transformation – Sorption – Biodegradation - Ion exchange - Precipitation - Hydrological consideration in landfill design - Ground water pollution.

UNIT-IV:
Waste Stabilization: Stabilization & Solidification of wastes - Micro and macro encapsulation -Absorption – Adsorption – Precipitation – Detoxification - Mechanism of stabilization - Organic and inorganic stabilization -utilization of solid waste for soil improvement - case studies.

UNIT-V:
Remediation of Contaminated Soils: Ex-situ and in-situ remediation – Solidification - bioremediation - incineration soil washing – phytoremediation - soil heating – vitrification -bioventing.

TEXT BOOKS:

1. Geo-Environmental Engineering, Hari D. Sharma, Krishna R. Reddy, John Wiley and Sons, 2004
2. Geotechnical Practice for Waste Disposal, B. E. Daniel, Chapman & Hall, 1993
3. Waste Disposal in Engineered Landfills, Manoj Datta, Narosa Publishing House, 1997

REFERENCES:

1. Industrial Solid Waste Management and Landfilling Practice, Manoj Datta, B. P. Parida, B. K. Guha, Narosa Publishing, 1999
2. Landfill Waste Pollution and Control, K. Westlake, Albion Publishing Ltd., 1995
3. Hazardous Waste Management, C. A. Wentz, McGraw-Hill, 1989
4. Environmental Geotechnology (Vol. I and II), Proceedings of the International Symposium on Environmental Publishing Company, 1986 and 1989
5. Environmental Indices, Theory and Practice, W. R. Ott, Ann Arbor, 1978

ONLINE RESOURCES:
1. https://nptel.ac.in/courses/105102160
2. https://onlinecourses.nptel.ac.in/noc19_ce37

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1CE415) ADVANCED SUB SURFACE INVESTIGATION AND INSTRUMENTATION

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:
- To develop clear idea about planning and execution of geotechnical investigation programme
- To apply the various methods of geotechnical investigation and the field tests based on field conditions
- To analyse and take proper engineering decisions in practical situations
- To gain the knowledge about the instrumentation for critical sites

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Examine subsurface investigation based on the requirement of civil engineering project and site condition
CO-2: Execute different subsurface exploration tests, collect disturbed/undisturbed samples for laboratory tests and can suggest design parameters
CO-3: Understand various methods for estimation of dynamic soil properties required for design purpose
CO-4: Develop instrumentation scheme for monitoring of critical sites

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	1	3	-	2	-	1	3	-	2	3	3	1	1
CO-2	3	3	3	3	1	-	-	2	3	1	2	2	2	3	2
CO-3	3	3	2	3	3	1	1	-	2	1	2	2	3	2	2
CO-4	3	3	3	3	3	-	2	1	3	1	1	3	3	3	2

UNIT-I:
Planning an Investigation Programmes: Factors to be considered; Exploration for preliminary and detailed design; Guidelines for location, depth and spacing of drilling bore holes.
Exploration Techniques: Methods of Boring, Accessible exploration and Semi-direct methods, Machinery used for drilling and applicable soil types, Stabilization of boreholes.

UNIT-II:
Sampling: Disturbed and undisturbed soil sampling, representative samples, Methods to minimize sample disturbance; Types of samplers, Preservation and handling of samples, Ground water observations, Borehole logging.

UNIT-III:

Field Tests: Standard Penetration Test, Dynamic and static cone penetration tests, Pressure meter test, Field vane shear, Field permeability test, Soil Investigation report.

UNIT-IV:

Instrumentation: Settlement gauges, Inclinometers, Stress measurements, Seismic measurements, and Pore pressure measurements.

UNIT-V:

Geophysical Methods: Geophysical methods-types, Seismic Methods – Electrical Resistivity Methods – Electrical Profiling Method – Seismic refraction method, Sub-soil Investigation Report

TEXT BOOKS:

1. Foundation Analysis and Design, J. E. Bowles, McGraw-Hill International Edition, 1997
2. Advanced Soil Mechanics, B. M. Das, 2nd Edition, Taylor and Francis, 1997
3. Soil Mechanics & Foundation Engineering, P. Purushothama Raj, Pearson Education India, 2008

REFERENCES:

1. Principles of Geotechnical Engineering, M. Das, 7th Edition, Cengage Learning, 2010
2. Ground Property Characterization from In-Situ Testing, M. D. Desai, GS-Surat Chapter, 2005
3. Geotechnical Engineering Investigation Manual, R. E. Hunt, 2nd Edition, McGraw-Hill, 2005
4. Soil Sampling, American Society of Civil Engineers, 1999
5. Engineering Properties of Soil and Their Measurements, B. Bowles, McGraw-Hill Companies, 1992

ONLINE RESOURCES:

1. https://archive.nptel.ac.in/courses/105/103/105103182
2. https://archive.nptel.ac.in/noc/courses/noc19/SEM1/noc19-ce10